Cardiovascular System (CARD) · NPLEX-P1-CARD-10

# CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-10-01-C01-BD1

## Brain dump 1 · foundational retrieval: Draw the lymphatic capillary and how lymph forms

A competent response contains

· Lymphatic capillaries, also called initial lymphatics, begin as blind-ended tubes near blood capillary beds.

· The endothelium is thin, the basal lamina is incomplete or absent, and there are no pericytes and no smooth muscle.

· Neighbouring endothelial cells overlap like roof shingles, held by loose discontinuous junctions, so the free inner edge of each overlap swings inward as a flap, the primary lymphatic valve.

· Anchoring filaments tie the outer surface of the endothelium to the collagen and elastic fibres of the surrounding tissue.

· Outward Starling forces are capillary hydrostatic pressure and interstitial oncotic pressure; inward forces are plasma oncotic pressure and interstitial hydrostatic pressure.

· The outward forces slightly exceed the inward ones, so a small net volume is filtered, and some albumin escapes with it that cannot be moved back into blood against its concentration gradient.

· Steps: net capillary filtration adds fluid and protein to the interstitium; the tissue swells and anchoring filaments pull the lymphatic capillary open; the overlapping flaps part and interstitial fluid enters; rising pressure inside the vessel closes the flaps; lymph moves on into valved collecting lymphatics.

· The vessel opens because swelling tissue pulls the anchoring filaments taut, so the capillary is pulled open rather than compressed; it closes because internal pressure presses the flaps shut, which makes flow one way.

· The gaps admit protein, lipoprotein, bacteria, and whole cells, which is the opposite of a continuous blood capillary built to retain them.

Distinctions that matter

· The flap valve of the lymphatic capillary is formed by overlapping endothelial cells; the bicuspid valves further along belong to collecting lymphatics.

· Tissue swelling opens a lymphatic capillary and compresses a blood capillary, and the anchoring filaments are the reason for the difference.

Common misconceptions

· Drawing a continuous basal lamina and tight junctions on a lymphatic capillary. Both are what blood capillaries use to keep protein in, and the lymphatic needs protein to get out.

· Saying filtration happens because plasma oncotic pressure is low. Plasma oncotic pressure is the main inward force, and net filtration is small because the outward forces exceed it only slightly.

Where it is taught: Section C01, opening paragraph, structure of the lymphatic capillary, and the formation of lymph flow diagram of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C01-BD2

## Brain dump 2 · integration and application: What stops working when protein cannot get back

A competent response contains

· Protein leaves the interstitium only by lymph, because no mechanism moves it back into blood against its concentration gradient, so a blocked drain leaves protein behind.

· Interstitial protein rises, interstitial oncotic pressure rises toward plasma levels, the gradient that holds fluid inside vessels is erased, and net filtration increases, so the limb swells even with normal capillary pressure.

· Lymph protein is about 2 g/dL from most tissues, about 6 g/dL from the liver, and about 3 to 4 g/dL from the intestine.

· Hepatic lymph is the most protein-rich because hepatic sinusoids leak albumin freely.

· A duct draining externally loses dietary long-chain fat and fat-soluble vitamins, albumin, and lymphocytes.

· Enterocytes package long-chain fatty acids as triglyceride in chylomicrons, which are too large for blood capillaries, so they enter the lacteal of each villus and make intestinal lymph milky, which is chyle.

· Most dietary fat therefore reaches blood through the thoracic duct and bypasses the liver on first pass, while medium-chain fatty acids are small enough to enter portal blood.

· No lymphatic capillaries are present in epidermis, cartilage, cornea, bone marrow, or brain and spinal cord parenchyma, which drain along perivascular routes to meningeal lymphatics.

Distinctions that matter

· Lymphedema fails at the drain while the other kinds of edema push too much fluid out, which is why the fluid here is protein-rich.

· Chylomicron size, not the type of fatty acid alone, decides which route absorbed fat takes.

Common misconceptions

· Saying macrophages or the venous end of the capillary clear interstitial protein. The lymphatic is the only route back.

· Expecting all absorbed fat to enter the portal vein. Long-chain fat in chylomicrons reaches the blood at the neck veins.

Where it is taught: Section C01, paragraphs on what the lymphatics return, and the final paragraph on tissues without lymphatics of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C01-RT01

## Retrieval exercise: Rebuild the steps that form lymph

A competent response contains

· Step 1: net capillary filtration adds fluid and protein to the interstitium.

· Step 2: the tissue swells and the anchoring filaments pull the lymphatic capillary open.

· Step 3: the overlapping endothelial flaps part and interstitial fluid enters.

· Step 4: rising pressure inside the vessel closes the flaps.

· Step 5: lymph moves on into valved collecting lymphatics.

· The anchoring filaments pull outward on the vessel wall as the surrounding tissue swells, so the vessel is held open instead of being squeezed shut.

· The flaps close because pressure inside the vessel now exceeds interstitial pressure and presses them against the overlap, so flow can only go forward.

· Resting lymph flow is about 120 mL per hour, about 100 mL of it through the thoracic duct, which is 2 to 3 L per day.

· That daily volume matches net filtration because a net filtration pressure of about 0.3 mmHg acting on a large filtration coefficient filters about 2 mL per minute, and whatever is not reabsorbed must leave as lymph.

Distinctions that matter

· Opening the vessel depends on forces outside it; closing it depends on pressure inside it.

· Flow through the thoracic duct is most of total lymph flow but not all of it, since the right lymphatic duct carries the rest.

Common misconceptions

· Drawing the tissue squeezing the lymphatic capillary closed as it swells. The anchoring filaments make swelling open it.

Where it is taught: Section C01, formation of lymph flow diagram and the paragraph on what the lymphatics return of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-BD1

## Brain dump 1 · foundational retrieval: Draw the thoracic duct from end to end

A competent response contains

· The cisterna chyli is a variable sac anterior to about L1 to L2 beside the aorta, formed where the right and left lumbar trunks and the intestinal trunk join.

· Lumbar trunks carry lymph from the lower limbs, pelvis, abdominal wall, kidneys, adrenals, and gonads after the iliac and lumbar nodes; the intestinal trunk carries lymph including chyle from the stomach, intestines, liver, pancreas, and spleen after the preaortic nodes.

· The duct ascends with the aorta through the aortic hiatus at about T12.

· In the posterior mediastinum it lies on the vertebral bodies behind the esophagus, with the aorta on its left and the azygos vein on its right.

· At about T4 to T6 it crosses to the left behind the esophagus and ascends along the left edge of the esophagus.

· At about C7 it arches laterally behind the left carotid sheath and ends at the left venous angle, the junction of the left internal jugular and left subclavian veins, where a terminal valve keeps blood out.

· Near its end it receives the left jugular, left subclavian, and often the left bronchomediastinal trunks.

· The right lymphatic duct, about 1 cm long, is formed by the right jugular, subclavian, and bronchomediastinal trunks, which may instead open into the veins separately, and ends at the right venous angle.

· It drains only the right head and neck, right upper limb, and right thorax including the right breast and right lung; the thoracic duct drains everything else.

· Much of the left lower lobe of the lung drains against that pattern, to right tracheobronchial nodes.

· The course and termination of both ducts vary between individuals.

Distinctions that matter

· The duct begins on the right at the cisterna chyli and ends on the left at the venous angle, and the crossing at about T4 to T6 is where that changes.

· Only four regions use the right lymphatic duct, so anything outside them returns by the thoracic duct.

Common misconceptions

· Drawing the duct in front of the esophagus. It lies behind the esophagus on the vertebral bodies.

· Expecting a right lymphatic duct in every body. The three right trunks often open into the veins separately.

Where it is taught: Section C02, paragraphs on trunks and the cisterna chyli, the course of the thoracic duct, and the right lymphatic duct of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-BD2

## Brain dump 2 · integration and application: Work backward from the node to the primary site

A competent response contains

· Left supraclavicular node: abdominal lymph ascends the thoracic duct to the left venous angle, so gastric carcinoma can appear there first, the Virchow node.

· Lumbar or para-aortic nodes near the renal vessels: testis or ovary, so a testicular or ovarian tumour.

· Superficial inguinal nodes: scrotum, vulva, anal canal below the pectinate line, and superficial lower limb, so a scrotal or vulval skin lesion or a low anal carcinoma.

· Axillary nodes: upper limb and lateral breast, which is most breast lymph, so a breast carcinoma or an arm infection.

· Popliteal nodes: lateral foot and posterolateral calf, draining onward to deep inguinal nodes, so a lesion of the lateral foot.

· The testes descend from the posterior abdominal wall and bring their lymphatics with them, so testicular cancer goes to para-aortic nodes, while scrotal skin is body wall and drains to superficial inguinal nodes.

· The anal canal above the pectinate line is hindgut and drains to internal iliac nodes, with the upper rectum going to inferior mesenteric nodes; below the pectinate line it is ectoderm and drains to superficial inguinal nodes.

· Most of the breast drains to axillary nodes and the medial part drains to parasternal nodes along the internal thoracic vessels.

· The medial hand and forearm drain first to cubital, also called supratrochlear, nodes before the axilla, so an elbow node can enlarge first.

· Each region drains to a predictable first node group, which is where infection causes tender lymphadenitis and where carcinoma usually metastasizes first.

Distinctions that matter

· Skin over an organ and the organ itself often drain to different node groups, which is the point of the scrotum against testis comparison.

· The pectinate line marks a change in embryological origin, and the node groups follow that origin rather than the position in the canal.

Common misconceptions

· Assuming a testicular tumour spreads to inguinal nodes because the testis sits in the scrotum. Lymphatics follow the path of descent, not the final position.

· Sending all breast lymph to the axilla. The medial part goes to parasternal nodes.

Where it is taught: Section C02, first node groups table and the paragraph on development and the classic findings of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-PR01

## Prediction exercise: Predict the effect of a block low in the thoracic duct

A competent response contains

· Lymph outflow from both lower limbs and the pelvis falls, because their lumbar trunks drain into the cisterna chyli below the block and have no other route onward, so lower body swelling can follow.

· Lymph outflow from the abdominal viscera, including intestinal chyle, falls for the same reason, and chylous ascites can follow because the obstructed chyle has nowhere to go.

· The left upper limb, left head and neck, and left thorax still drain normally, because the left jugular, left subclavian, and left bronchomediastinal trunks join the duct near its end, above the block.

· The right head and neck, right upper limb, and right thorax still drain normally through the right lymphatic duct, which is a separate route to the right venous angle.

· Pressure inside the duct below the obstruction rises, and the intrinsic pump can generate about 50 to 100 mmHg against an obstruction.

· The obstructed vessels dilate, so the valve leaflets no longer meet, lymph refluxes between contractions, and forward flow falls even though the muscle is still contracting.

Distinctions that matter

· Where a trunk joins the duct relative to the block decides whether its region is affected, so the level of the block matters more than the size of the region.

· A high pump pressure and a failing pump are not a contradiction: pressure rises first, and valve incompetence from dilation is what finally defeats it.

Common misconceptions

· Expecting the whole body to stop draining because the thoracic duct carries most lymph. Everything joining above the block, and everything using the right duct, is unaffected.

· Assuming an obstructed lymphatic simply stops. It pumps harder first, and fails when dilation separates the valve leaflets.

Where it is taught: Section C02, stop and predict box, with the valve and intrinsic pump paragraph in Section C03 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-BD1

## Brain dump 1 · foundational retrieval: Lay out the whole lymphatic pump

A competent response contains

· Lymph begins in an interstitium slightly below atmospheric pressure, about -3 mmHg in subcutaneous tissue, and ends in veins a few mmHg above atmospheric, in the legs against gravity.

· Valves allow flow in only one direction, so any compression of a valved segment pushes lymph toward the veins rather than back and forth.

· The flap valves of the lymphatic capillaries are the first set; collecting lymphatics carry bicuspid endothelial valves every few millimetres, and the segment between two valves is a lymphangion.

· Cycle: lymph enters through the open upstream valve; filling stretches the smooth muscle of the wall; stretch triggers contraction and pressure rises; the upstream valve closes and the downstream valve opens; lymph is ejected and stretches the next lymphangion; the emptied segment relaxes and refills from upstream.

· Stretch from filling increases the rate and force of contraction, so the pump works harder when more lymph arrives, much as stretch increases cardiac stroke volume through the Frank-Starling mechanism.

· Sympathetic stimulation increases the activity, endothelial nitric oxide relaxes the muscle between contractions, and inflammation depresses it.

· Extrinsic pumps: skeletal muscle contraction, passive movement, pulsation of neighbouring arteries since deep lymphatics share arterial sheaths, and external compression.

· Exercise can raise lymph flow about 10 to 30 times, and immobility lowers it.

· During inspiration the descending diaphragm raises abdominal pressure around the cisterna chyli while intrapleural pressure falls around the thoracic duct, so lymph is drawn upward, and valves block backflow during expiration.

Distinctions that matter

· The intrinsic pump is the lymphangion's own smooth muscle; the extrinsic pumps are everything outside the vessel that compresses it.

· Valves do not move lymph, they decide the direction of whatever force is applied.

Common misconceptions

· Looking for a central pump for lymph. There is none, and the lymphangions plus external compression do the work.

· Treating breathing as a thoracic effect only. It works because abdominal pressure rises while intrathoracic pressure falls at the same moment.

Where it is taught: Section C03, opening paragraph, valves and the intrinsic pump, the lymphangion pump cycle, and extrinsic pumps of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-BD2

## Brain dump 2 · integration and application: Why the pump runs out of room

A competent response contains

· Subcutaneous interstitial fluid pressure is normally about -3 mmHg, largely because the lymphatics keep pumping fluid out.

· As interstitial pressure rises from about -6 mmHg to slightly above 0 mmHg, lymph flow increases about twentyfold.

· Above about 1 to 2 mmHg flow plateaus, because the tissue pressure that fills the lymphatic capillaries also compresses the larger collecting vessels.

· Once filtration exceeds that maximum flow, edema accumulates.

· Safety factor one: low compliance of the negative-pressure interstitium, so a small fluid gain raises interstitial pressure steeply, worth about 3 mmHg.

· Safety factor two: lymph flow can rise 10 to 50 times, worth about 7 mmHg.

· Safety factor three: that extra flow washes protein out faster than it leaks in, lowering interstitial oncotic pressure, worth about 7 mmHg, and it depends on the second factor existing.

· Together they let capillary pressure rise about 17 mmHg before significant edema appears.

· Removing the regional nodes and vessels loses the second and third factors at once, so leaking protein accumulates and interstitial oncotic pressure rises instead of falling, and the tissue gains protein-rich fluid even with a normal capillary pressure.

· Lymph finally empties against central venous pressure, so right heart failure and superior vena cava obstruction both slow lymph outflow and raise capillary filtration at the same time, which contributes to effusions and ascites.

Distinctions that matter

· The same rise in tissue pressure that fills the initial lymphatics compresses the collecting vessels, which is why the curve has a ceiling rather than rising without limit.

· The third safety factor is a consequence of the second, so losing lymph flow removes both at once.

Common misconceptions

· Expecting lymph flow to keep rising with tissue pressure. It plateaus at about 1 to 2 mmHg, and edema is what happens past that point.

· Treating the safety factors as independent. Protein washout exists only while lymph flow can increase.

Where it is taught: Section C03, paragraph on interstitial pressure and outflow pressure, and the worked example on safety factors of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-RT01

## Retrieval exercise: Rebuild the lymphangion pump cycle

A competent response contains

· Step 1: lymph enters through the open upstream valve.

· Step 2: filling stretches the smooth muscle of the wall.

· Step 3: stretch triggers contraction and pressure rises.

· Step 4: the upstream valve closes and the downstream valve opens.

· Step 5: lymph is ejected and stretches the next lymphangion.

· Step 6: the emptied segment relaxes and refills from upstream.

· The contraction was triggered by stretch, and the stretch came from lymph filling the segment, so output rises with filling.

· The ejected lymph passes into the next lymphangion downstream and stretches it, which triggers that segment to contract in turn, so contraction runs in sequence toward the veins.

· An obstructed thoracic duct can reach about 50 to 100 mmHg.

· If a chronically obstructed vessel dilates, the valve leaflets no longer meet, lymph refluxes between contractions, and the pump fails.

Distinctions that matter

· Only one valve of a lymphangion is open at any step, which is what makes the segment a pump rather than a tube.

· Pump failure in chronic obstruction comes from valve incompetence after dilation, not from weak muscle.

Common misconceptions

· Drawing both valves open during contraction. The upstream valve must close before lymph can be ejected forward.

· Assuming a dilated lymphatic moves more lymph because it holds more. Dilation separates the leaflets and lets lymph run backward.

Where it is taught: Section C03, the lymphangion pump cycle and the valves and intrinsic pump paragraph of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-BD1

## Brain dump 1 · foundational retrieval: Build the lymphoid organs from their compartments

A competent response contains

· Primary lymphoid organs are bone marrow, where B cells mature, and thymus, where T cells mature; lymphocytes there generate antigen receptors and are tested against self.

· Secondary lymphoid organs are lymph nodes, which sample lymph, spleen, which samples blood, and mucosa-associated lymphoid tissue, which samples mucosal surfaces; naive lymphocytes meet antigen there and recirculating lymphocytes pass through each.

· Lymph node: afferent lymph enters the subcapsular sinus, percolates through macrophage-lined sinuses, and leaves at the hilum through efferent vessels; B cells sit in cortical follicles drawn by CXCL13 acting on CXCR5 around follicular dendritic cells displaying intact antigen; germinal centres hold proliferating B cells where activation-induced cytidine deaminase drives somatic hypermutation and class switching and higher-affinity cells are selected by antigen and follicular helper T cells; the paracortex holds T cells, dendritic cells, and high endothelial venules through which naive lymphocytes enter using L-selectin and CCR7; medullary cords hold plasma cells secreting antibody.

· Follicular hyperplasia reflects a B-cell response and paracortical hyperplasia a T-cell response, often viral.

· Spleen: white pulp has T cells sheathing the central arteriole in the periarteriolar lymphoid sheath beside B-cell follicles, with a marginal zone that traps blood-borne antigen; red pulp is where blood empties into splenic cords and must squeeze through narrow slits between sinusoidal endothelial cells, so stiff, aged, or antibody-coated red cells are removed and inclusions are pitted; the red pulp holds about one third of platelets; the spleen has no afferent lymphatics.

· Thymus: lies in the anterior superior mediastinum, develops from third pharyngeal pouch endoderm, and involutes into fat after puberty; each lobule has a dark cortex packed with thymocytes behind a blood-thymus barrier and a pale medulla containing Hassall corpuscles, which are whorls of keratinized epithelial cells; it has no follicles and no afferent lymphatics.

· MALT: Peyer patches, largest in the ileum, plus appendix, bronchial tissue, and tonsils, which have a partial capsule; it is unencapsulated; M cells transcytose luminal antigen inward to the underlying lymphoid cells; B cells there switch to IgA under TGF-beta, home back to mucosa through alpha4 beta7 integrin binding MAdCAM-1, and secrete dimeric IgA carried across the epithelium by the polymeric immunoglobulin receptor.

· T-cell maturation: a marrow precursor enters near the corticomedullary junction; a double-negative cell rearranges TCR beta; pre-TCR signalling makes it double positive; TCR alpha rearrangement completes the alpha beta receptor; in the cortex weak binding to self MHC survives while no binding dies by neglect; class II binding keeps CD4 and class I binding keeps CD8; in the medulla strong binding to AIRE-driven self antigen causes deletion or a regulatory fate; naive, MHC-restricted, self-tolerant T cells leave.

· The first six steps are cortical, the strong-binding deletion step is medullary, and cortical positive selection makes T cells useful while medullary negative selection makes them safe.

· In the marrow, pro-B cells rearrange heavy chain genes with RAG1 and RAG2, a functional heavy chain with a surrogate light chain forms the pre-B cell receptor, and after light chain rearrangement an immature B cell whose IgM binds self strongly undergoes receptor editing and then deletion if editing fails; survivors mature in the spleen into naive B cells bearing IgM and IgD.

Distinctions that matter

· Nodes have afferent lymphatics and spleen and thymus do not, which is why nodes sample lymph while the spleen samples blood.

· Positive selection is about usefulness and happens in the cortex; negative selection is about safety and happens in the medulla.

Common misconceptions

· Drawing follicles in the thymus. The thymus has cortex and medulla and no follicles at all.

· Placing plasma cells in the germinal centre. Antibody-secreting plasma cells sit in the medullary cords.

Where it is taught: Section C04, opening paragraphs, the thymus paragraph, the T-cell maturation flow diagram, the node, spleen, and MALT paragraphs, and the compartments table of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-BD2

## Brain dump 2 · integration and application: Read a biopsy and a blood film back to a compartment

A competent response contains

· Child one has follicles and germinal centres but an empty paracortex, so T cells are missing; the spleen's periarteriolar lymphoid sheath would also be empty while its follicles and marginal zone are intact; this fits DiGeorge syndrome from the 22q11.2 deletion with a hypoplastic thymus.

· Child two has a normal paracortex but no follicles, no germinal centres, and no plasma cells in the medullary cords, so B cells are missing; the spleen's follicles and marginal zone would be empty while the periarteriolar sheath is intact; this fits X-linked, also called Bruton, agammaglobulinemia.

· Serum antibody of every class is low in child two because the plasma cells of the medullary cords are the source of secreted antibody, and without B cells there are no plasma cells to make any isotype.

· Child three has an enlarged paracortex with normal follicles, which is paracortical hyperplasia, a T-cell response and usually viral.

· In the thymic cortex a T-cell receptor must bind self peptide and self MHC weakly to survive, since no binding dies by neglect; in the medulla it must avoid binding AIRE-driven self antigen strongly, since strong binding causes deletion or diversion to regulatory T cells.

· Child one has lost both of those processes, and the organ in which they are lost is the thymus.

· After splenectomy: Howell-Jolly bodies, thrombocytosis, and overwhelming sepsis from encapsulated bacteria.

· The three organisms are Streptococcus pneumoniae, Haemophilus influenzae type b, and Neisseria meningitidis.

· The red cell findings appear because red pulp cords and the narrow sinusoidal slits are where inclusions are pitted and aged or coated cells are culled, and thrombocytosis follows because the red pulp normally holds about one third of platelets.

· The infection risk follows because encapsulated organisms need splenic clearance and marginal zone IgM, which is T-independent antibody against polysaccharide.

· AIRE mutation in the thymic medulla causes autoimmune polyendocrine syndrome type 1, with hypoparathyroidism, adrenal insufficiency, and chronic mucocutaneous candidiasis.

Distinctions that matter

· An immunodeficiency that removes one cell type empties that cell's compartment in every lymphoid organ, so reading a node tells you what the spleen will look like.

· Hyperplasia of a compartment and depletion of a compartment both point to the same cell line, one responding and one absent.

Common misconceptions

· Reading absent germinal centres as a T-cell problem because follicular helper T cells are needed for them. Here the paracortex and T-cell numbers are normal and the B cells themselves are missing.

· Attributing post-splenectomy infection risk to loss of all antibody. The gap is in clearance of encapsulated organisms and in marginal zone IgM against polysaccharide.

Where it is taught: Section C04, the node paragraph, the spleen paragraph, the thymus paragraph, the T-cell maturation flow diagram, and the compartments table of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-BD1

## Brain dump 1 · foundational retrieval: Causes of lymphedema and how it hardens

A competent response contains

· Too much filtration produces edema through raised capillary pressure, low plasma oncotic pressure, or increased capillary permeability.

· In lymphedema filtration is normal and the drain fails; because protein leaves the interstitium only by lymph, the retained fluid is rich in protein, and secondary causes far outnumber primary ones.

· Lymph node dissection removes nodes and the vessels running through them: arm after axillary dissection, leg after pelvic or inguinal dissection.

· Radiation causes fibrosis that narrows and obliterates vessels, often years later, in axilla, pelvis, or groin, especially alongside surgery.

· Tumour blocks vessels and nodes with tumour emboli, in metastatic carcinoma and lymphoma, and gives peau d'orange over a breast carcinoma.

· Filariasis: adult worms dilate lymphatics and make valves incompetent, and inflammation and infection scar them, affecting legs, scrotum, and breast in endemic tropical regions.

· Primary developmental causes leave too few or malfunctioning vessels or valves: Milroy disease from VEGFR3 mutation, FOXC2 mutation, and Turner syndrome.

· The organisms are Wuchereria bancrofti and Brugia species; adult worms live in the lymphatics for years while microfilariae circulate in blood, typically at night.

· Vessel damage comes from living worms dilating the lymphatics, from inflammation provoked by dying worms and their Wolbachia endosymbionts, and from repeated bacterial infection adding scarring; the result is leg lymphedema progressing to elephantiasis, hydrocele, and sometimes chyluria.

· Cycle: lymphatic outflow falls; protein-rich fluid accumulates and holds water; stagnant protein and lipid provoke chronic inflammation; fibroblasts deposit collagen and fat expands; fibrosis damages the remaining lymphatics and cellulitis recurs; each infection and increment of fibrosis reduces outflow further.

· At the fibrosis step, chronic inflammation with CD4 T cells and macrophages leads TGF-beta to stimulate fibroblasts to lay down collagen, and lymph stasis promotes fat growth.

· Early lymphedema is soft and pitting and improves with elevation, although its high protein content holds water so swelling returns when the limb hangs down; later, fibrosis and fat do not move under pressure, so the limb is firm and nonpitting and no longer shrinks with elevation, and the skin thickens with hyperkeratosis and papillomatous change.

· Inability to pinch a skin fold at the base of the second toe is the Stemmer sign.

· Lost immune cell traffic and protein-rich fluid favour recurrent streptococcal cellulitis, and lymphangiosarcoma is a rare late complication, classically in the arm years after mastectomy with axillary dissection.

Distinctions that matter

· Every cause of lymphedema removes, obstructs, or disables lymphatic pathways, and the downstream course is the same whichever one it was.

· Pitting against nonpitting here is a matter of time rather than of cause, because fibrosis and fat take months to years to develop.

Common misconceptions

· Explaining nonpitting late lymphedema by the protein content of the fluid. Protein holds water but still pits; what does not move under a finger is collagen and fat.

· Expecting filariasis to damage lymphatics only through the worms themselves. Inflammation from dying worms and repeated bacterial infection do much of the scarring.

Where it is taught: Section C05, opening paragraph, the causes of lymphedema table, the filariasis paragraph, the paragraph on pitting to nonpitting swelling, and the self-perpetuating cycle of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-BD2

## Brain dump 2 · integration and application: Reason out a milky effusion

A competent response contains

· The thoracic duct has been injured, and a left-sided effusion places the injury above the crossing at about T4 to T6, because above the crossing the duct runs along the left edge of the esophagus while below it lies behind the esophagus to the right of midline.

· Pleural triglyceride above about 110 mg/dL supports chylothorax, below about 50 mg/dL makes it unlikely, and chylomicrons in the fluid confirm it.

· Fasting lymph is clear and scant, so the fluid was thin while the patient was not eating; eating fat sends chylomicrons through the duct and turns the fluid milky.

· Chyle stays turbid after centrifugation, which separates it from the fluid of an infected pleural space, where turbidity settles out.

· Pseudochylothorax, a chronic cholesterol effusion inside a thickened pleura, has cholesterol above about 200 mg/dL and no chylomicrons, so measuring cholesterol and looking for chylomicrons settles that comparison, and an acute onset argues against it as well.

· A continuing leak depletes albumin, lymphocytes, dietary fat, and fat-soluble vitamins, because the duct normally returns escaped plasma protein, carries recirculating lymphocytes from the nodes, and delivers absorbed long-chain fat as chylomicrons along with the vitamins dissolved in it.

· Surgical trauma and malignancy, especially lymphoma, are the leading causes of chylothorax.

Distinctions that matter

· The side of a chylous effusion locates the injury relative to the T4 to T6 crossing, which is the one piece of anatomy the question turns on.

· Turbidity that persists after centrifugation means suspended chylomicrons, while turbidity from cells and debris settles.

Common misconceptions

· Taking a milky effusion as proof of chyle without a triglyceride level. A chronic cholesterol effusion also looks milky and has no chylomicrons.

· Expecting a duct injury to leak the same fluid whatever the patient eats. Fat intake is what loads the duct with chylomicrons.

Where it is taught: Section C05, the chylothorax paragraph and the worked example on a milky effusion after esophagectomy of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-CP01

## Comparison table: Three kinds of swelling that look alike

A competent response contains

· What accumulates: lymphedema holds protein-rich interstitial fluid that is later replaced by collagen and fat; heart failure or low albumin edema holds low-protein interstitial fluid; myxedema holds glycosaminoglycan rather than free fluid.

· Protein content: high in lymphedema because protein leaves only by lymph, low in the filtration edemas, and not applicable in myxedema since it is not free fluid.

· Mechanism: lymphedema is failure of the drain with normal filtration; heart failure raises capillary hydrostatic pressure and low albumin lowers plasma oncotic pressure, both raising filtration; myxedema is deposition in the tissue rather than a Starling problem.

· Pitting: lymphedema pits early and becomes nonpitting once fibrosis and fat develop, because those do not move under pressure; filtration edema stays pitting because it is free fluid; myxedema is nonpitting because it is glycosaminoglycan.

· Elevation: early lymphedema improves with elevation and returns when the limb hangs down because its protein holds water, late lymphedema does not improve at all, filtration edema improves with elevation, and myxedema does not change.

· Skin over time: in lymphedema the skin thickens with hyperkeratosis and papillomatous change and cellulitis recurs; in filtration edema the skin is stretched and shiny without that thickening; in myxedema the tissue is doughy and does not indent.

· Pointer: the Stemmer sign, inability to pinch a skin fold at the base of the second toe, points to lymphedema; a raised jugular venous pressure or a low serum albumin points to the filtration edemas; nonpitting swelling without a drainage history points to myxedema.

Distinctions that matter

· Two kinds of swelling here are nonpitting for different reasons: lymphedema because fibrosis and fat have replaced the fluid, myxedema because the material was never free fluid.

· Protein content is the single feature that separates lymphedema from every filtration edema, and everything that follows comes from it.

Common misconceptions

· Treating nonpitting as a single finding with one cause. It has to be read with the history and the skin.

· Calling lymphedema a high-pressure problem. Capillary pressure is normal and the drain has failed.

Where it is taught: Section C05, opening paragraph and the paragraph on pitting to nonpitting swelling of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-CA01

## Clinical application: A firm, swollen arm four years after breast surgery

A competent response contains

· Recognize: secondary lymphedema of the right arm. The axillary lymph node dissection removed the nodes and the vessels running through them, and the axillary radiation caused fibrosis that narrowed and obliterated the remaining vessels, which is why the swelling began months after surgery and progressed over years.

· Explain: protein leaves the interstitium only by lymph, so a failed drain leaves protein behind and the retained fluid is protein-rich. Interstitial oncotic pressure then rises instead of falling, the gradient holding fluid inside vessels is eroded, and the tissue gains fluid even though capillary pressure, plasma albumin, and capillary permeability are all normal.

· Differentiate: heart failure edema is a raised capillary pressure problem, is low in protein, stays pitting, and would be bilateral with a raised jugular venous pressure, which is normal here; low albumin edema is also low in protein and pitting and would be generalized, and her albumin is 4.1 g/dL. This arm is unilateral, firm, nonpitting, with thickened skin and a positive Stemmer sign, on the side of the dissection and radiation.

· Investigate: the normal jugular venous pressure and normal heart sounds exclude a raised central venous pressure, the serum albumin of 4.1 g/dL excludes hypoalbuminemia, the normal left arm confirms a regional rather than a systemic cause, and the skin fold at the base of the second finger that cannot be pinched up is the Stemmer sign applied to the hand, which points to lymphatic failure.

· Decide: retained protein and lipid drove chronic inflammation with CD4 T cells and macrophages, TGF-beta stimulated fibroblasts to lay down collagen, and lymph stasis promoted fat growth. Collagen and fat now occupy the tissue, and neither moves under a finger nor drains with elevation.

· Justify: the safety factors against edema are a stiff negative-pressure interstitium worth about 3 mmHg, a 10 to 50 fold rise in lymph flow worth about 7 mmHg, and protein washout by that extra flow worth about 7 mmHg. Removing the nodes and vessels loses the second and third at once, so about 14 mmHg of the roughly 17 mmHg of protection is gone and interstitial protein rises rather than falling. Each episode of cellulitis then enters the cycle: lost immune cell traffic and protein-rich fluid invite streptococcal infection, infection and fibrosis damage the lymphatics that remain, outflow falls further, and the next episode starts from a worse baseline.

Distinctions that matter

· Surgery and radiation are separate insults with different timing, one immediate and one developing over years, and this history contains both.

· The change from pitting to nonpitting marks a change in what is in the tissue, not a change in how severe the drainage failure is.

Common misconceptions

· Reading a unilateral swollen arm as venous obstruction by default. The dissection and radiation history, the protein-rich picture, and the Stemmer sign point to the lymphatics.

· Treating recurrent cellulitis as a complication that sits outside the mechanism. It is part of the cycle that makes the lymphedema progress.

Where it is taught: Section C05, opening paragraph, the causes of lymphedema table, the paragraph on pitting to nonpitting swelling, and the self-perpetuating cycle, with the safety factors in Section C03 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-Q01

Board-style question: Which compartment the organ lacks

Correct answer: C. Arrest of B-cell maturation in the bone marrow

Why it is right: B cells populate the cortical follicles and germinal centres and become the plasma cells of the medullary cords, so an arrest of B-cell maturation in the marrow empties all three of those compartments while leaving the paracortex and its high endothelial venules normal, and with no plasma cells every antibody class is low.

A is wrong: Failed positive selection in the thymic cortex would empty the paracortex and the splenic periarteriolar lymphoid sheath and lower circulating T cells, but this boy's paracortex and T-cell count are normal.

B is wrong: Failed entry through high endothelial venules would leave the paracortex poorly populated as well, since that is the route naive lymphocytes of both lines take out of the blood, and the paracortex here is normal.

D is wrong: Absent M cells would impair sampling of luminal antigen at mucosal surfaces, which affects mucosal IgA responses rather than removing follicles, germinal centres, and plasma cells from a lymph node.

A competent response contains

· Absent follicles, absent germinal centres, and absent plasma cells in the medullary cords, with a normal paracortex and normal T-cell numbers, place the defect in the B-cell line.

· Plasma cells in the medullary cords are the source of secreted antibody, so their absence explains IgG, IgA, and IgM all being low, and X-linked agammaglobulinemia is the arrest of B-cell maturation in the marrow.

Distinctions that matter

· An immunodeficiency empties the compartment belonging to the missing cell type in every lymphoid organ, so a node biopsy localizes the defect to a cell line.

Common misconceptions

· Reading absent germinal centres as a T-cell defect. Germinal centres need follicular helper T cells, but here the T-cell compartment is intact and the B cells are missing.

Where it is taught: Section C04 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-Q02

Board-style question: Where a testicular tumour spreads first

Correct answer: A. Lumbar (para-aortic) nodes near the renal vessels

Why it is right: The testes develop on the posterior abdominal wall and drag their lymphatic vessels with them as they descend, so the first node group for a testicular tumour is the lumbar or para-aortic group near the renal vessels.

B is wrong: Superficial inguinal nodes drain scrotal skin, vulva, the anal canal below the pectinate line, and the superficial lower limb, and the scrotal skin here is normal.

C is wrong: Internal iliac nodes drain pelvic viscera and the anal canal above the pectinate line, not the testis.

D is wrong: Popliteal nodes drain the lateral foot and posterolateral calf and pass lymph onward to deep inguinal nodes, so they have no role in testicular drainage.

A competent response contains

· The testis descends from the posterior abdominal wall and brings its lymphatics with it, so testicular lymph drains to lumbar, also called para-aortic, nodes near the renal vessels.

· Scrotal skin is body wall and drains to superficial inguinal nodes, which is why the normal overlying skin matters in this case.

Distinctions that matter

· Lymphatic drainage follows the path of embryological descent rather than the adult position of the organ.

Common misconceptions

· Sending testicular lymph to inguinal nodes because the testis sits in the scrotum. The organ and the skin over it drain to different node groups.

Where it is taught: Section C02 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.